

ABHINAV SINGHAL

Principal Software Engineer

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EXPERIENCE

Principal Engineer /Product Owner

Dell Technologies

📅 12/2023 - Present

- Design and develop distributed file system components for Dell Power Scale (One FS), working closely with Free BSD file system internals.
- Enhanced Dell Power Scale (One FS) architecture by delivering **Quality of Service (QoS)** and **enterprise data deletion** capabilities, helping customers achieve predictable performance and simplifying data management.
- Drive the scalability, reliability, and performance of core storage data-path components, with a focus on journaling, distributed consensus, and data lifecycle management.
- **Evaluated and integrated GitHub Spec Kit into the development lifecycle**, establishing a specification-first workflow that improved requirement clarity, design traceability, and AI-assisted implementation quality.
- Developed **Kubernetes Custom Resource Definitions (CRDs)** and corresponding **Custom Resources** to represent application-specific configuration and desired state.
- Designed and implemented custom **Kubernetes controllers** to reconcile Custom Resources with the desired state and automate application lifecycle management.
- Lead technical initiatives, drive architectural decisions, design reviews, and mentoring engineers on storage and systems programming best practices.

Staff Engineer

Ruckus Networks

📅 12/2021 - 12/2023

- Staff engineer in an Individual Contributor role with Ruckus Networks for their patented cloud SaaS offering (DPSK).
- Building and maintaining a set of 15 distributed microservices using Spring boot and Spring Kafka.
- Converted current REST-based implementation to Event-Driven Architecture using Confluent Kafka and spring cloud stream.
- Managed and optimized deployment of distributed microservices on GCP/GKE/Kubernetes, reducing the DPSK deployment footprint from **15 microservices to 6**, lowering deployment complexity and operational cost.(Confirmed insights after running kubecost analysis).
- Designed and deployed cloud-native applications on **Google Kubernetes Engine (GKE)** using Kubernetes CLI (kubectl) and **Terraform**.
- Worked with **Kubernetes HPA and KEDA** for application autoscaling, including migration from resource-based HPA to queue/workload-based scaling.
- Feature Enhancements - Helped in scaling the application to support 75K passphrases per tenant against 20K initially, revoke-unrevoked access, policy configuration and other core features.

SUMMARY

Principal Software Engineer with 12+ years of expertise specializing in file systems, distributed architecture and event-driven design. Skilled in Linux kernel development, file system design, and container orchestration (Kubernetes, CSI). Notable for reducing microservice deployment complexity and increasing application throughput. Committed to advancing enterprise infrastructure through collaborative leadership and technical excellence.

KEY ACHIEVEMENTS



Boosted Storage Efficiency

Implemented and delivered Quality of Service based on read/write latencies. Enhanced throttling metrics.



Enhanced Cloud Scalability

Scaled DPSK capacity to support 75k passphrases per tenant.



Optimized Deployment Costs

Reduced microservices in DPSK from 15 to 6, cutting deployment and debugging costs on Google Kubernetes Engine.

EDUCATION

B. Tech in Computer Science

Jaypee Institute of Information Technology, Noida

%
82 / 100

SKILLS

Distributed Systems

Distributed File system

Spec-Driven Development

Event-Driven Architecture

Kubernetes

Java

Golang

Kafka

Spring boot

Spring Cloud

Spring cloud stream

GCP/GKE

Convergent Mediation

BRIM

IBM MQ/ Rabbit MQ

AI Harness

EXPERIENCE

IT Technology Consultant

SAP Labs

📅 11/2018 - 11/2021

- Worked as a Java developer in an individual contributor role. Helped develop and formulate a middleware orchestration engine developed and managed on the SAP Cloud Platform. (Pivotal Cloud Foundry). The project was built on microservice architecture, which included the following components - Spring boot, Spring Data JPA, Spring Cloud (Service discovery, Service authentication (XSUUA tokens), Resiliency), and HANA Cloud
 - Submitted proof of concept (POC) to the business for predicting the credit scores/credit trends using the past credit ratings using the HANA PAL (Predictive Analysis Library) and various python modules (NumPy, pandas, scikit learn)
 - Successfully designed an end-to-end application using CDS annotations and standard Fiori UI elements, which allowed businesses to monitor the metrics including business reports, customer segregation within regions, public-private buyers' information, and various other KPIs
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Senior Application Developer

Accenture Services Pvt. Ltd

📅 08/2014 - 11/2018

- Responsible for delivering change-requests/functional requirements for a leading telecom company based out of Saudi Arabia
- Understanding client requirements and breaking them into design-level documents (HLD LLD), thus creating a roadmap for implementing them into low-level code functionality
- Maintained the best JAVA coding practices by creating and maintaining SONAR code quality rules
- Submitted Proof of Concept (POC) establishing the process to test the reliability of message queues by integrating the IBM MQ with SOAP UI
- Improved the efficiency of the team by enabling early identification of bugs/defects in the applications using SONAR code quality custom review rules, and Reducing manual unit testing efforts by implementing Mocker Frameworks-based tools in JAVA

CERTIFICATIONS

**Level 1 certification for Work soft Certify:
Web Test Automation**

Mediation Zone Foundational Training

Digital Route